

Week 1 Questions

Multiple Choice Questions (MCQs)

1. What is the primary goal of an agent in Reinforcement Learning?

- a) To mimic the training data as closely as possible.
- b) To find patterns in unlabeled data.
- c) To maximize reward in a particular situation.
- d) To classify data into pre-defined categories.

2. How does Reinforcement Learning primarily differ from Supervised Learning?

- a) RL uses labeled data, while supervised learning does not.
- b) RL learns from outcomes and feedback, while supervised learns from a training set with correct answers.
- c) RL is only used for robotics, while supervised learning is used for all other tasks.
- d) RL requires a teacher, while supervised learning is self-taught.

3. According to the document, what is a key characteristic of the data used in Reinforcement Learning?

- a) It is always structured data like housing examples.
- b) It is not part of the input like in supervised or unsupervised learning; it is obtained through interaction.
- c) It must be un-structured data like images and audio.
- d) It comes from a static dataset provided at the beginning.

4. Which of the following is listed as an application of Reinforcement Learning?

- a) Filtering spam emails from an inbox.
- b) A master chess player planning moves and anticipating replies.
- c) Identifying objects in a static image.

d) Clustering customers into market segments.

5. What determines the "best solution" in a Reinforcement Learning model?

- a) The one with the lowest computational cost.
- b) The one that is most similar to the initial state.
- c) The one that is decided based on the maximum reward.
- d) The one that follows a pre-programmed path exactly.

6. Which of the following is mentioned as an advantage of Reinforcement Learning?

- a) It requires very little data and computation.
- b) It is the best approach for solving simple, straightforward problems.
- c) It can handle non-deterministic environments where outcomes are not always predictable.
- d) The agent's behavior is always easy to debug and interpret.

7. What is a major disadvantage of Reinforcement Learning mentioned in the text?

- a) It cannot be combined with other techniques like deep learning.
- b) Its performance is highly dependent on the quality of the reward function.
- c) It can only be used in robotics.
- d) It never makes errors, so no correction is needed.

8. In the Reinforcement Learning process, after the model takes an action, what does it receive?

- a) A new static dataset to train on.
- b) A label for the input data.
- c) Feedback that helps determine if the choice was correct, neutral, or incorrect.

d) A pre-programmed command for the next move.

9. The document states that Reinforcement Learning is the science of what?

- a) Pattern Recognition
- b) Data Clustering
- c) Decision Making
- d) Image Processing

10. Why is Reinforcement Learning not preferable for solving simple problems?

- a) Because it is not accurate enough for simple tasks.
- b) Because it needs a lot of data and computation, which is overkill for simple problems.
- c) Because simple problems always require a teacher.
- d) Because it cannot handle structured data.

True/False Questions

1. In Reinforcement Learning, the training data has the answer key, similar to supervised learning.

2. Reinforcement Learning is useful for automated systems that need to make many small decisions without human guidance.

3. The training in Reinforcement Learning is based solely on the initial input, and the model does not learn after the first output.

4. A key advantage of RL is that it can only be used in deterministic environments where the outcome of an action is always predictable.

- 5. Reinforcement Learning uses algorithms that learn from outcomes to decide which action to take next.**
- 6. According to the document, an example of an RL application is an adaptive controller adjusting a petroleum refinery's operation in real-time.**
- 7. One of the disadvantages of RL is that it is always easy to debug and interpret why the agent is behaving a certain way.**
- 8. The input for an RL problem should be a final goal state from which the model will not deviate.**
- 9. Reinforcement learning is a flexible approach that can be combined with other machine learning techniques, such as deep learning.**
- 10. In a chess example, the choice of move is informed by planning and anticipating possible replies, which is characteristic of Reinforcement Learning.**

Week 2 & 3 Questions

Multiple Choice Questions (MCQs)

1. According to the document, what is the most widely used deep learning method for image processing?

- a) Deep Belief Nets (DBN)
- b) Stacked Auto-encoders (SAE)
- c) Convolutional Neural Networks (CNNs)
- d) Deep Boltzmann Machines (DBM)

2. What explicit assumption do CNN architectures make about the input data?

- a) The data is sequential.
- b) The data is numerical.
- c) The inputs are images.
- d) The inputs are pre-classified.

3. What is the primary function of the convolutional layer in a CNN?

- a) To reduce the dimensionality of the data.
- b) To perform the final classification.
- c) To extract features from the input data using filters.
- d) To flatten the data into a one-dimensional vector.

4. What problem does the convolution operation solve when working with images?

- a) It increases the number of free parameters.
- b) It reduces the number of free parameters, allowing for deeper networks.
- c) It eliminates the need for an activation function.
- d) It converts the image to grayscale.

5. If you apply a single 5x5 filter to an image, approximately how many learnable parameters are required, regardless of the image size?

- a) 5
- b) 10
- c) 25
- d) 100

6. Which type of pooling layer uses the maximum value from each cluster of neurons in the prior layer?

- a) Sum pooling
- b) Average pooling
- c) Min pooling
- d) Max pooling

7. Which of the following is NOT mentioned as a frequently used CNN architecture trained on ImageNet data?

- a) AlexNet
- b) VGGNet
- c) ResNet-101
- d) GANet

8. The two main parts of a CNN model, as described in the document, are:

- a) Feature Learning and Classification
- b) Input Layer and Output Layer
- c) Training and Testing
- d) Forward Propagation and Backpropagation

9. What is the most commonly used activation function in CNNs, which helps eliminate negative values?

- a) Sigmoid
- b) Tanh
- c) ReLU
- d) Softmax

10. The parameter that determines the step size by which a filter moves across an image is called:

- a) Padding
- b) Stride
- c) Kernel
- d) Channel

11. What is the purpose of adding zero padding to an input image before convolution?

- a) To increase the image's resolution.
- b) To add noise to the image for better training.
- c) To control the spatial size of the output volume.
- d) To convert the image to a binary format.

12. What is the primary purpose of the pooling layer?

- a) To apply the activation function.
- b) To reduce the spatial size (dimensionality) of the representation.
- c) To increase the number of parameters.
- d) To connect every neuron to every other neuron.

13. After feature extraction, what is done to the multi-dimensional output before it is fed into the fully connected layers?

- a) It is converted back into an image.
- b) It is flattened into a single vector.
- c) It is passed through another convolutional layer.
- d) It is discarded.

14. The SoftMax layer at the end of a CNN is used to:

- a) Apply a non-linearity to the features.
- b) Down-sample the feature maps.
- c) Translate the final output into a vector of probabilities for each class.
- d) Initialize the weights of the network.

15. In the provided CNN architecture table, what is the output size after the first conv_1 layer?

- a) 32x32x3
- b) 32x32x32
- c) 15x15x32
- d) 7x7x64

16. According to the document, what does a fully connected layer do?

- a) It applies a convolution operation.
- b) It connects all the input feature maps from the previous layer to a larger feature map or vector.
- c) It performs max-pooling on the input.
- d) It adds zero padding to the input.

17. The formula for calculating the dimension of the pooled part in max pooling is given as $\dim \text{ of } P = ((m1 + 2p - n1)/s) \times \dots$. In this formula, what do $n1 \times n2$ represent?

- a) The dimensions of the input image.
- b) The dimensions of the padding kernel.
- c) The stride value.
- d) The number of filters.

18. What is the range of values in the output vector produced by a SoftMax layer?

- a) -1 to 1
- b) 0 to 1, and they all sum to 1
- c) 0 to infinity
- d) -infinity to infinity

19. According to the document, how does a CNN with ReLU compare to one with a hyperbolic tangent activation function?

- a) It trains much slower.
- b) It trains much faster.
- c) It is less accurate.
- d) It requires more data.

20. In a convolutional layer, what is the filter also known as?

- a) The stride
- b) The padding
- c) The kernel function
- d) The feature map

True/False Questions

- 1. Deep learning is an older learning algorithm that has been replaced by newer methods.**
- 2. Convolutional Neural Networks are made up of neurons that have learnable weights and biases.**
- 3. A fully connected layer for a small 100 x 100 image would only need 100 weights for each neuron in the second layer.**
- 4. In a convolutional layer, the ReLU activation function is typically applied before the convolution operation.**
- 5. Average pooling uses the maximum value from each of a cluster of neurons.**
- 6. AlexNet, VGGNet, and ResNet-101 are examples of classic CNN models trained on the ImageNet dataset.**
- 7. The feature learning part of a CNN involves the convolutional and pooling layers.**
- 8. The kernel (or filter) in a convolutional layer is a non-linear function used to extract features.**
- 9. Padding is a technique where rows and columns of zeros are added to the outside of an input image.**

10. A bias term is not added in a convolutional layer; only the weights from the filter are used.

11. The pooling layer increases the spatial size of the convoluted features to get a more detailed image.

12. After feature extraction, the multi-dimensional feature map is flattened into a single vector to be used as input for the fully connected layers.

13. The SoftMax function ensures that the output probabilities for all classes sum to 1.

14. In a CNN, the convolutional and pooling layers can only be repeated once to avoid overfitting.

15. CNNs are only useful for image classification and cannot be used for other tasks like speech recognition.

16. The formula $[n, n, nC] * [f, f, nC]$ describes the convolution of a 3D image with a single filter that has the same depth (nC).

17. The last fully connected layer in a CNN provides a score for each object, which can be converted into a probability.

18. The example in the document shows that an input vector $[1, -2, -1]$ passed through a SoftMax layer results in the highest probability being for the class represented by the second element.

19. The architecture table in the document shows that the network uses both max pooling and average pooling layers.

20. The output from a fully connected layer is always the final class label, with no further activation function needed.



Week 5 Questions

Multiple Choice Questions (MCQs)

1. What distinguishes RNNs from traditional deep learning networks?

- a) They do not use backpropagation
- b) They assume inputs and outputs are independent
- c) Their output depends on prior elements in a sequence
- d) They cannot process sequential data

2. Which of the following is an application of RNNs mentioned in the PDF?

- a) Image compression
- b) Signal processing
- c) Database indexing
- d) Graph algorithms

3. What does the hidden state in an RNN act as?

- a) A fixed weight matrix
- b) A memory storing information about previous inputs
- c) The final output layer
- d) A dropout regularizer

4. Which idiom is used in the PDF to explain RNNs?

- a) "Break a leg"
- b) "Feeling under the weather"
- c) "Piece of cake"
- d) "Spill the beans"

5. What problem do RNNs suffer from during backpropagation?

- a) Underfitting
- b) Vanishing and exploding gradients
- c) Data scarcity
- d) Over-regularization

6. Which architecture is proposed as a solution to the vanishing gradient problem?

- a) GRU
- b) CNN
- c) LSTM
- d) Autoencoder

7. How many gates does an LSTM cell have according to the PDF?

- a) 1
- b) 2
- c) 3
- d) 4

8. What are the three gates in an LSTM cell?

- a) Read gate, write gate, execute gate
- b) Input gate, output gate, forget gate
- c) Forward gate, backward gate, reset gate
- d) Memory gate, update gate, flush gate

9. What creates a feedback loop in an RNN?

- a) Passing the output to the loss function
- b) Passing the hidden state from one time step to the next
- c) Reversing the input sequence
- d) Using only one activation function

10. In LSTM, what might be excluded from the cell state if a gender pronoun like "she" appears multiple times?

- a) The pronoun itself
- b) The forget gate
- c) Unnecessary gender information
- d) The entire sentence

11. Traditional deep learning networks assume inputs and outputs are:

- a) Sequential
- b) Dependent on each other
- c) Independent of each other
- d) Random

12. Which field is mentioned as benefiting from RNNs for classification and regression on large signal datasets?

- a) Bioformatics
- b) Real-time prediction
- c) Image rendering
- d) Quantum computing

True / False Questions

1.RNNs use training data to learn.

2.In RNNs, the hidden state is reset at every time step.

3.LSTM networks have 4 gates in each cell.

4.The idiom "feeling under the weather" can be understood regardless of word order according to the PDF.

5.Vanishing gradients cause training failures in deep neural networks.

6.Exploding gradients make gradients too small during backpropagation.

7.LSTM stands for Long Short-Term Memory.

8.Signal processing is mentioned as an application of RNNs.

9.LSTM was designed to worsen the vanishing gradient problem.

10. The forget gate in LSTM controls which information to discard from the cell state.

11. Traditional deep learning networks are better than RNNs for sequential data because they assume dependency between inputs.

12. RNNs are used for text analytics tasks like sentiment analysis.

13. The hidden state in an RNN is passed from one time step to the next to create a feedback loop.

Week 6 & 7 Questions

Multiple Choice Questions (MCQs)

1. What does generative AI primarily do?

- A) Make predictions about existing data
- B) Create new content based on learned patterns
- C) Store large datasets
- D) Analyze sentiment in text

2. Which of the following is an example of a generative AI model for images?

- A) ChatGPT
- B) GitHub Copilot
- C) DALL-E
- D) Jukebox

3. What is the first phase in how generative AI works?

- A) Tuning
- B) Generation
- C) Evaluation
- D) Training

4. What architecture introduced in 2017 by Google is used to develop large language models?

- A) Convolutional Neural Network
- B) Recurrent Neural Network
- C) Transformer
- D) Autoencoder

5. What does an LLM use to predict the next word in a sentence?

- A) Human-like understanding
- B) Probability and pattern matching
- C) Exact memorization
- D) Random guessing

6. Which Python library is used to analyze text sentiment?

- A) Transformers
- B) TextBlob
- C) NLTK
- D) spaCy

7. What does the polarity score in TextBlob sentiment analysis indicate?

- A) Length of the text
- B) Emotion type (anger, joy)
- C) Sentiment from -1 to 1
- D) Number of words

8. What kind of data is used during the training phase of a foundation model?

- A) Only labeled data
- B) Only numerical data
- C) Raw, unstructured, unlabeled data
- D) Only images

9. Which of the following is a multimodal foundation model capable of supporting several kinds of content generation?

- A) GPT-3
- B) BERT
- C) Multimodal foundation models
- D) TextBlob

10. What is the purpose of the attention map in a transformer?

- A) To store tokens
- B) To capture relationships between tokens
- C) To delete irrelevant words
- D) To count words

11. Which pipeline is used in the Transformers library for text summarization?

- A) text-generation
- B) sentiment-analysis
- C) summarization
- D) chatbot

12. What does the PDF say about whether an LLM "thinks" like a human?

- A) Yes, LLMs think exactly like humans
- B) No, LLMs simulate understanding using probability
- C) LLMs only memorize answers
- D) LLMs have emotions

13. Which company created ChatGPT?

- A) Google
- B) Meta
- C) Anthropic
- D) OpenAI

True / False Questions

1. Generative AI can only create text content.

2. The training phase of generative AI uses huge volumes of raw, unlabeled data.

- 3. A transformer encodes each word as a vector, not a token.**
- 4. In TextBlob, a polarity of 0 means positive sentiment.**
- 5. GitHub Copilot is an example of generative AI for music.**
- 6. The summarization pipeline in Transformers requires pip installing transformers.**
- 7. The attention map helps the transformer understand context when generating new text.**
- 8. An LLM is trained only on images and videos.**
- 9. The sentiment analysis function in the PDF returns "Positive" if polarity is greater than 0.**
- 10. Jukebox is an example of a generative AI model for music.**
- 11. The generative AI systems learn to generate objects that look like the data they were trained on.**